

Parallel & Perpendicular Lines — Solutions

Complete Step-by-Step Solutions with Trap Alerts & Key Insights

Question 1 [Easy]

Correct Answer: A) 4

Step-by-Step Solution:

- Parallel lines have equal slopes.
- Slope of line $j = 4$, so any parallel line also has slope 4.

Why each wrong option fails:

- B) -4 — This is the negative, used for perpendicular opposite signs only.
- C) $1/4$ — This is the reciprocal, not the same slope.
- D) $-1/4$ — This is the negative reciprocal, which is for perpendicular lines.

■ ■ **TRAP ALERT:** Students mix up the parallel rule (same slope) with the perpendicular rule (negative reciprocal).

■ **KEY INSIGHT:** Parallel = same slope. Perpendicular = negative reciprocal.

■ **FAST METHOD:** Parallel → same slope. Answer: 4.

■ **VERDICT:** Hand-solve — pure definition recall.

Question 2 [Easy]

Correct Answer: C) $5/2$

Step-by-Step Solution:

- Perpendicular slope = negative reciprocal of $-2/5$.
- Flip: $-5/2$. Negate: $5/2$.
- Perpendicular slope = $5/2$.

Why each wrong option fails:

- A) $2/5$ — Just flipped the sign, didn't take reciprocal.
- B) $-2/5$ — Same slope (parallel, not perpendicular).
- D) $-5/2$ — Only took reciprocal without negating.

■ ■ **TRAP ALERT:** Two operations needed: flip AND negate. Doing only one gives wrong answers.

■ **KEY INSIGHT:** Negative reciprocal means flip the fraction AND change the sign.

■ **FAST METHOD:** Flip $-2/5$ to $-5/2$, then negate to $5/2$.

■ **VERDICT:** Hand-solve — definition-based.

Question 3 [Easy]

Correct Answer: B) $y = 3x + 12$

Step-by-Step Solution:

- Parallel lines share the same slope.
- Slope of $y = 3x - 7$ is 3.
- Only $y = 3x + 12$ has slope 3.

Why each wrong option fails:

- A) $y = -3x + 7$ — Slope is -3 (opposite sign).
- C) $y = (1/3)x - 7$ — Slope is 1/3 (reciprocal).
- D) $y = -3x - 7$ — Slope is -3.

■ TRAP ALERT: Same y-intercept (-7) does NOT mean parallel. Same SLOPE does.

■ KEY INSIGHT: For parallel, ONLY the slope matters. The y-intercept can be anything.

■ FAST METHOD: Match the coefficient of x. Only option B has 3x.

■ VERDICT: Hand-solve — instant visual scan of slopes.

Question 4 [Medium]

Correct Answer: C) $y = (1/3)x + 2$

Step-by-Step Solution:

- First find slope of $6x + 2y = 10$:
- $2y = -6x + 10$
- $y = -3x + 5$
- Slope = -3.
- Perpendicular slope = negative reciprocal of -3 = 1/3.
- Only $y = (1/3)x + 2$ has slope 1/3.

Why each wrong option fails:

- A) $y = -3x + 5$ — Same slope (parallel, not perpendicular).
- B) $y = 3x - 1$ — Negated the slope but didn't take reciprocal.
- D) $y = (-1/3)x + 4$ — Took reciprocal but didn't negate. Product is +1, not -1.

■ TRAP ALERT: Option D is the most tempting — it has 1/3 but with the wrong sign.

■ KEY INSIGHT: Perpendicular slopes multiply to -1. Check: $(-3)(1/3) = -1$ ✓.

■ FAST METHOD: Slope = -3 → perp slope = 1/3. Scan options for 1/3 positive.

■ VERDICT: Hand-solve — convert to slope-intercept, then match.

Question 5 [Medium]

Correct Answer: A) $y = -4x + 13$

Step-by-Step Solution:

- Parallel to $y = -4x + 1$ means slope = -4.
- Point-slope: $y - 5 = -4(x - 2)$
- $y - 5 = -4x + 8$
- $y = -4x + 13$

Why each wrong option fails:

- B) $y = -4x - 3$ — Subtracted 8 instead of adding.
- C) $y = (1/4)x + 13$ — Used perpendicular slope instead of parallel.
- D) $y = 4x + 13$ — Dropped the negative sign from the slope.

■ **TRAP ALERT:** Using the perpendicular slope (1/4) instead of the parallel slope (-4).

■ **KEY INSIGHT:** Parallel = same slope as the given line. Just find the new y-intercept.

■ **FAST METHOD:** $b = 5 - (-4)(2) = 5 + 8 = 13$. So $y = -4x + 13$.

■ **VERDICT:** Hand-solve — quick substitution.

Question 6 [Medium]

Correct Answer: A) (0, 10)

Step-by-Step Solution:

- Slope of $y = (2/3)x + 5$ is $2/3$.
- Perpendicular slope = $-3/2$.
- Line n: $y - 1 = -3/2(x - 6)$
- $y - 1 = -3/2 x + 9$
- $y = -3/2 x + 10$
- y-intercept = (0, 10).

Why each wrong option fails:

- B) (0, -8) — Sign error in point-slope expansion.
- C) (0, 7) — Used slope $2/3$ instead of $-3/2$.
- D) (0, -10) — Sign error in the y-intercept.

■ **TRAP ALERT:** Forgetting to negate when finding the perpendicular slope.

■ **KEY INSIGHT:** Perpendicular to $2/3$ is $-3/2$ (flip and negate).

■ **FAST METHOD:** $b = 1 - (-3/2)(6) = 1 + 9 = 10$.

■ **VERDICT:** Hand-solve — straightforward perpendicular slope + point.

Question 7 [Medium]

Correct Answer: B) $y = 2x + 7$

Step-by-Step Solution:

- Convert: $8x - 4y = 20 \rightarrow -4y = -8x + 20 \rightarrow y = 2x - 5$.
- Slope = 2. Parallel line also has slope 2.
- Only $y = 2x + 7$ has slope 2.

Why each wrong option fails:

- A) $y = -2x + 3$ — Slope is -2 (opposite sign).
- C) $y = (1/2)x - 5$ — Reciprocal slope.
- D) $y = -1/2 x + 1$ — Negative reciprocal (perpendicular).

■ ■ **TRAP ALERT:** The same y-intercept (-5) in option C might tempt students who confuse y-intercept with slope.

■ **KEY INSIGHT:** Convert to slope-intercept first, then match slopes.

■ **FAST METHOD:** Slope = $-A/B = -8/(-4) = 2$. Find the option with slope 2.

■ 🏠 **VERDICT:** Hand-solve — use -A/B shortcut.

Question 8 [Medium]

Correct Answer: A) $y = 3x - 4$

Step-by-Step Solution:

- Line v: $3x + 9y = 27 \rightarrow 9y = -3x + 27 \rightarrow y = -(1/3)x + 3$.
- Slope of v = $-1/3$.
- Perpendicular slope = 3.
- Line w passes through (0, -4) with slope 3: $y = 3x - 4$.

Why each wrong option fails:

- B) $y = -3x - 4$ — Negated without taking reciprocal.
- C) $y = (1/3)x - 4$ — Only took reciprocal, didn't negate.
- D) $y = -(1/3)x - 4$ — Same slope as line v (parallel).

■ ■ **TRAP ALERT:** The perpendicular slope of $-1/3$ is 3 (not $1/3$ or -3).

■ **KEY INSIGHT:** $(-1/3)(3) = -1$ ✓ confirms perpendicularity.

■ **FAST METHOD:** Given (0, -4) is the y-intercept, just determine slope. Perp to $-1/3 = 3$.

■ 🏠 **VERDICT:** Hand-solve — y-intercept is directly given, making this very quick.

Question 9 [Medium]

Correct Answer: $-1/4$

Step-by-Step Solution:

- Convert $2x + 8y = 16$ to slope-intercept:
- $8y = -2x + 16$
- $y = -(1/4)x + 2$
- Slope = $-1/4$.
- For parallel lines, $k = -1/4$.

■ ■ **TRAP ALERT:** Students may write $k = 1/4$ (positive) by dropping the negative from $-2x/8$.

■ **KEY INSIGHT:** Parallel means same slope. Just find the slope of the second line.

■ **FAST METHOD:** Slope = $-A/B = -2/8 = -1/4$. So $k = -1/4$.

■ 🏠 **VERDICT:** Hand-solve — slope shortcut.

Question 10 [Medium]

Correct Answer: A) $x = 3$

Step-by-Step Solution:

- $y = 5$ is a horizontal line (slope = 0).
- A line perpendicular to a horizontal line is vertical.
- Vertical line through (3, -2) is $x = 3$.

Why each wrong option fails:

- B) $y = -2$ — Horizontal, same type as $y = 5$ (parallel, not perpendicular).
- C) $x = -2$ — Vertical but uses y -coordinate instead of x -coordinate.
- D) $y = 5x - 2$ — Has a defined slope, not vertical.

■ ■ **TRAP ALERT:** Using the y -coordinate (-2) instead of the x -coordinate (3) for the vertical line equation.

■ **KEY INSIGHT:** Horizontal ■ Vertical. $y = c$ ■ $x = c$.

■ **FAST METHOD:** Perpendicular to horizontal = vertical. $x = 3$.

■ 🏠 **VERDICT:** Hand-solve — conceptual, no calculation.

Question 11 [Medium]

Correct Answer: C) $y = 3$

Step-by-Step Solution:

- $(4, -1)$ and $(4, 7)$ share $x = 4$, so this is a vertical line.
- Perpendicular to a vertical line is a horizontal line.
- $y = 3$ is horizontal.

Why each wrong option fails:

- A) $x = 2$ — Vertical (parallel, not perpendicular).
- B) $y = 4x - 1$ — Has a defined slope (not horizontal).
- D) $x = -1$ — Vertical (parallel).

■ TRAP ALERT: Picking another vertical line ($x = \text{something}$) instead of a horizontal one.

■ KEY INSIGHT: Vertical ■ Horizontal. Recognize vertical lines by identical x -coordinates.

■ FAST METHOD: Same x -coords = vertical line. Perp = horizontal = $y = \text{constant}$.

■ VERDICT: Hand-solve — recognize the vertical/horizontal relationship.

Question 12 [Medium]

Correct Answer: A) -2

Step-by-Step Solution:

- Line a: $5x - 10y = 30 \rightarrow -10y = -5x + 30 \rightarrow y = (1/2)x - 3$.
- Slope of a = $1/2$.
- Perpendicular slope = -2 .
- So $m = -2$.

Why each wrong option fails:

- B) $-1/2$ — Only negated the slope.
- C) $1/2$ — Same slope (parallel).
- D) 2 — Only took reciprocal, didn't negate.

■ TRAP ALERT: Option D (2) results from taking the reciprocal without negating.

■ KEY INSIGHT: $(1/2)(-2) = -1$ ✓ confirms perpendicularity.

■ FAST METHOD: Slope = $1/2 \rightarrow \text{perp} = -2$. Instant.

■ VERDICT: Hand-solve — basic reciprocal operation.

Question 13 [Medium]

Correct Answer: A) $y = -(4/3)x + 5$

Step-by-Step Solution:

- Perpendicular to slope $3/4$ gives slope $= -4/3$.
- Using point $(3, 1)$: $y - 1 = -(4/3)(x - 3)$
- $y - 1 = -(4/3)x + 4$
- $y = -(4/3)x + 5$

Why each wrong option fails:

- B) $y = (4/3)x + 5$ — Forgot to negate.
- C) $y = -(4/3)x - 3$ — Arithmetic error in y-intercept.
- D) $y = (3/4)x + 5$ — Same slope as original (parallel).

■ ■ **TRAP ALERT:** Using the original slope $3/4$ instead of the perpendicular $-4/3$.

■ **KEY INSIGHT:** Perpendicular to $3/4$ is $-4/3$. Remember: flip AND negate.

■ **FAST METHOD:** $b = 1 - (-4/3)(3) = 1 + 4 = 5$. $y = -(4/3)x + 5$.

■ 🏠 **VERDICT:** Hand-solve — standard process.

Question 14 [Easy]

Correct Answer: B) The lines are perpendicular.

Step-by-Step Solution:

- The condition $m_{\blacksquare} \cdot m_{\blacksquare} = -1$ is the definition of perpendicular slopes.
- This means the two lines are perpendicular.

Why each wrong option fails:

- A) Parallel — Parallel means $m_{\blacksquare} = m_{\blacksquare}$, not $m_{\blacksquare} \cdot m_{\blacksquare} = -1$.
- C) Intersect at origin — Perpendicular lines can intersect anywhere.
- D) Same y-intercept — Unrelated to the slope product.

■ ■ **TRAP ALERT:** Confusing the perpendicular condition (product $= -1$) with the parallel condition (equal slopes).

■ **KEY INSIGHT:** $m_{\blacksquare} \cdot m_{\blacksquare} = -1$ is the algebraic definition of perpendicular lines.

■ **FAST METHOD:** Definition recall: product of slopes $= -1$ means perpendicular.

■ 🏠 **VERDICT:** Hand-solve — pure concept.

Question 15 [Hard]

Correct Answer: 2

Step-by-Step Solution:

- Slope of $2x - 7y = 14$: slope = $-A/B = -2/(-7) = 2/7$.
- Parallel line also has slope $2/7$.
- Passes through $(-7, 0)$: $y - 0 = (2/7)(x - (-7))$
- $y = (2/7)(x + 7) = (2/7)x + 2$
- y-intercept = $(0, 2)$, so the value is 2.

■ TRAP ALERT: Using the constant 14 from the original equation as the y-intercept.

■ KEY INSIGHT: The y-intercept of a parallel line depends on the new point, not the original equation's constant.

■ FAST METHOD: $b = 0 - (2/7)(-7) = 0 + 2 = 2$.

■ VERDICT: Hand-solve — clean arithmetic with the given point.

Question 16 [Hard]

Correct Answer: A) $(-3/2, 0)$

Step-by-Step Solution:

- $3x + 4y = 24$ has slope = $-3/4$.
- Perpendicular slope = negative reciprocal = $4/3$.
- Line h through $(3, 6)$: $y - 6 = (4/3)(x - 3)$.
- $y = (4/3)x - 4 + 6 = (4/3)x + 2$.
- x-intercept: set $y = 0$: $(4/3)x + 2 = 0$.
- $(4/3)x = -2$, so $x = -2$ times $3/4 = -3/2$.
- The x-intercept is $(-3/2, 0)$.

Why each wrong option fails:

- B) $(3/2, 0)$ — Dropped the negative sign when solving.
- C) $(-6, 0)$ — Used the original slope $(-3/4)$ instead of perpendicular $(4/3)$.
- D) $(6, 0)$ — Used the x-coordinate of the given point.

■ TRAP ALERT: Forgetting the negative when solving $(4/3)x = -2$ gives the positive reciprocal $(3/2, 0)$.

■ KEY INSIGHT: Find perpendicular slope, write point-slope equation, then set $y = 0$ for the x-intercept.

■ FAST METHOD: Perp slope = $4/3$. Equation through $(3,6)$: $y = (4/3)x + 2$. Set $y = 0$: $x = -3/2$.

■ VERDICT: Hand-solve — systematic perpendicular slope then x-intercept.

Question 17 [Hard]

Correct Answer: A) $x = 4$

Step-by-Step Solution:

- PQ lies along the x-axis ($y = 0$) from (0,0) to (8,0). It is horizontal.
- The altitude from R to PQ must be perpendicular to PQ.
- Perpendicular to a horizontal line is a vertical line.
- Vertical line through R(4, 6) is $x = 4$.

Why each wrong option fails:

- B) $y = 6$ — Horizontal line through R, parallel to PQ (not perpendicular).
- C) $y = (2/3)x$ — Line from P to R, not the altitude.
- D) $x = 6$ — Uses y-coordinate instead of x-coordinate.

■ ■ TRAP ALERT: Confusing the altitude (perpendicular to a side) with a median or the side itself.

■ KEY INSIGHT: Altitude = perpendicular from a vertex to the opposite side.

■ FAST METHOD: PQ is horizontal $\rightarrow$ altitude is vertical $\rightarrow x = (\text{x-coord of R}) = 4$.

■ 🏠 VERDICT: Hand-solve — geometric reasoning, no computation.

Question 18 [Hard]

Correct Answer: A) $(0, 17/2)$

Step-by-Step Solution:

- Slope of s through (1,3) and (5,11): $m = (11-3)/(5-1) = 8/4 = 2$.
- Perpendicular slope = $-1/2$.
- Midpoint of segment: $((1+5)/2, (3+11)/2) = (3, 7)$.
- Line u through (3, 7) with slope $-1/2$:
- $y - 7 = -(1/2)(x - 3) \rightarrow y = -(1/2)x + 3/2 + 7 = -(1/2)x + 17/2$.
- y-intercept = $(0, 17/2)$.

Why each wrong option fails:

- B) (0, 8) — Used the y-coordinate of the midpoint as the y-intercept.
- C) $(0, 15/2)$ — Arithmetic error when combining $3/2 + 7$.
- D) $(0, 29/4)$ — Used wrong perpendicular slope.

■ ■ TRAP ALERT: Must find the perpendicular slope AND use the midpoint, not an endpoint.

■ KEY INSIGHT: This is the perpendicular bisector of the segment.

■ FAST METHOD: $b = 7 - (-1/2)(3) = 7 + 3/2 = 17/2$.

■ 🏠 VERDICT: Desmos — plot both points, find midpoint, verify the perpendicular line equation.

Question 19 [Hard]

Correct Answer: 1

Step-by-Step Solution:

- Slopes are $(a + 2)/3$ and $3/(a - 4)$. Perpendicular means product = -1.
- $[(a + 2)/3] \text{ times } [3/(a - 4)] = -1$.
- Simplify the left side: $3(a + 2) / [3(a - 4)] = (a + 2)/(a - 4)$.
- Set equal to -1: $(a + 2)/(a - 4) = -1$.
- Cross-multiply: $a + 2 = -(a - 4) = -a + 4$.
- Solve: $2a = 2$, so $a = 1$.
- Verify: slope $c = (1 + 2)/3 = 1$. Slope $d = 3/(1 - 4) = -1$. Product = -1. Confirmed.

■ TRAP ALERT: Cross-multiplying the perpendicular condition requires careful sign handling.

■ KEY INSIGHT: Set the slope product = -1, cross-multiply, solve the linear equation for a.

■ FAST METHOD: Product = -1: $(a + 2)/(a - 4) = -1$. Then $a + 2 = -a + 4$, giving $a = 1$.

■ VERDICT: Hand-solve — cross-multiply and solve.

Question 20 [Hard]

Correct Answer: C) 6

Step-by-Step Solution:

- Slope of $4x + 3y = 12$: slope = $-4/3$.
- Parallel line also has slope $-4/3$.
- Using points (3, 8) and (p, 4):
- $(4 - 8)/(p - 3) = -4/3$
- $-4/(p - 3) = -4/3$
- $p - 3 = 3$
- $p = 6$

Why each wrong option fails:

- A) 0 — Set up the slope equation incorrectly.
- B) 3 — Ignored the slope condition entirely.
- D) 9 — Used positive $4/3$ instead of $-4/3$.

■ TRAP ALERT: Swapping the order of subtraction in the slope formula leads to wrong signs.

■ KEY INSIGHT: If slope = $-4/3$ and rise = -4, then run must be 3. So $p = 3 + 3 = 6$.

■ FAST METHOD: Rise = -4, slope = $-4/3$, so run = 3. $p = 3 + 3 = 6$.

■ VERDICT: Hand-solve — rise/run reasoning.

Question 21 [Medium]

Correct Answer: B) 5

Step-by-Step Solution:

- $kx - 3y = 9 \rightarrow y = (k/3)x - 3$. Slope = $k/3$.
- Parallel to $y = (5/3)x + 2$ means slope = $5/3$.
- $k/3 = 5/3 \rightarrow k = 5$.

Why each wrong option fails:

- A) 3 — Confused k with the denominator.
- C) -3 — Sign error.
- D) -5 — Used negative of the correct answer.

■ ■ TRAP ALERT: Forgetting to account for the division by -3 (but here it's -(-3) so $k/3$).

■ KEY INSIGHT: Standard form $Ax + By = C$ has slope = $-A/B$. Here $-k/(-3) = k/3$.

■ FAST METHOD: $k/3 = 5/3 \rightarrow k = 5$. One step.

■ 🏠 VERDICT: Hand-solve — direct comparison.

Question 22 [Medium]

Correct Answer: A) $y = (1/2)x + 9/2$

Step-by-Step Solution:

- Slope through (0,0) and (3,-6): $(-6-0)/(3-0) = -2$.
- Perpendicular slope = $1/2$.
- Through (-1, 4): $y - 4 = (1/2)(x + 1)$
- $y = (1/2)x + 1/2 + 4 = (1/2)x + 9/2$

Why each wrong option fails:

- B) $y = (1/2)x + 4$ — Forgot the $+1/2$ from distributing.
- C) $y = -2x + 2$ — Used the original slope, not perpendicular.
- D) $y = 2x + 6$ — Used positive 2 (not the perpendicular slope).

■ ■ TRAP ALERT: Distributing $1/2$ into $(x+1)$ gives $1/2 x + 1/2$, not $1/2 x + 1$.

■ KEY INSIGHT: Find slope of given line $\rightarrow$ perpendicular slope $\rightarrow$ point-slope form.

■ FAST METHOD: $b = 4 - (1/2)(-1) = 4 + 1/2 = 9/2$.

■ 🏠 VERDICT: Hand-solve — systematic process.

Question 23 [Hard]

Correct Answer: A) $y = -(1/3)x + 17/3$

Step-by-Step Solution:

- Perpendicular to slope 3 means slope = $-1/3$.
- Through (2, 5): $y - 5 = -(1/3)(x - 2)$
- $y = -(1/3)x + 2/3 + 5 = -(1/3)x + 2/3 + 15/3 = -(1/3)x + 17/3$

Why each wrong option fails:

- B) $y = 3x + 5$ — Parallel, not perpendicular.
- C) $y = (1/3)x + 13/3$ — Wrong sign on slope.
- D) $y = -(1/3)x - 17/3$ — Sign error on y-intercept.

■ ■ **TRAP ALERT:** Converting 5 to 15/3 for fraction addition is where errors happen.

■ **KEY INSIGHT:** Always verify by plugging (2, 5) back into your answer.

■ **FAST METHOD:** $b = 5 + (1/3)(2) = 5 + 2/3 = 17/3$. Wait: $b = 5 - (-1/3)(2) = 5 + 2/3 = 17/3$.

■ 🏠 **VERDICT:** Hand-solve — standard process; Desmos for verification.

Question 24 [Hard]

Correct Answer: 5/8

Step-by-Step Solution:

- $4x - 6y = 3$ has slope = $-4/(-6) = 2/3$.
- Perpendicular slope = $-3/2$.
- $ax + by = 1$ has slope = $-a/b$, so $-a/b = -3/2$, meaning $a/b = 3/2$.
- Let $a = 3t$, $b = 2t$ for some t .
- Through (2, 1): $3t(2) + 2t(1) = 1 \rightarrow 6t + 2t = 1 \rightarrow 8t = 1 \rightarrow t = 1/8$.
- $a = 3/8$, $b = 2/8 = 1/4$.
- $a + b = 3/8 + 1/4 = 3/8 + 2/8 = 5/8$.

■ ■ **TRAP ALERT:** Two conditions give two unknowns. Use the slope ratio to relate a and b , then the point to find specific values.

■ **KEY INSIGHT:** Perpendicular slope + point on line = system of equations for a and b .

■ **FAST METHOD:** $a = 3t$, $b = 2t$. Point: $8t = 1 \rightarrow t = 1/8$. Then $a + b = 5t = 5/8$.

■ 🏠 **VERDICT:** Hand-solve — systematic algebraic approach.

Question 25 [Medium]

Correct Answer: B) $y = -2x + 9$

Step-by-Step Solution:

- Adjacent sides of a rectangle are perpendicular.
- Slope of $y = (1/2)x + 3$ is $1/2$.
- Perpendicular slope = -2 .
- Only $y = -2x + 9$ has slope -2 .

Why each wrong option fails:

- A) $y = (1/2)x - 4$ — Parallel (same slope), would be opposite side.
- C) $y = 2x + 1$ — Reciprocal without negating.
- D) $y = -(1/2)x + 3$ — Only negated, didn't take reciprocal.

■■ TRAP ALERT: Choosing the parallel line (same slope) instead of the perpendicular one.

■ KEY INSIGHT: Adjacent sides of a rectangle are perpendicular. Opposite sides are parallel.

■ FAST METHOD: Adjacent = perpendicular. Slope $1/2 \rightarrow$ perp slope = -2 . Match.

■ 🏠 VERDICT: Hand-solve — conceptual + slope matching.

■ TOPIC SUMMARY

1. **PARALLEL LINES:** Same slope, different y-intercepts. $m_1 = m_2$.
2. **PERPENDICULAR LINES:** Slopes are negative reciprocals. $m_1 \cdot m_2 = -1$.
3. **FINDING PERPENDICULAR SLOPE:** Flip the fraction AND change the sign. Two operations.
4. **STANDARD FORM SLOPE:** $Ax + By = C$ has slope $= -A/B$. Use this shortcut to avoid full conversion.
5. **VERTICAL ■ HORIZONTAL:** $x = c$ is perpendicular to $y = c$. These are special cases.
6. **RECTANGLE SIDES:** Adjacent sides of a rectangle are perpendicular. Opposite sides are parallel.
7. **ALTITUDE OF A TRIANGLE:** An altitude is perpendicular to the opposite side.
8. **PERPENDICULAR BISECTOR:** Find midpoint + perpendicular slope through it.
9. **COMMON TRAPS:** Only negating (not flipping), only flipping (not negating), using parallel slope for perpendicular.
10. **DESMOS vs HAND:** Hand-solve for slope identification and simple equations. Desmos for perpendicular bisectors and intersection points with messy fractions.